Logarithmic and Exponential Functions

Section 1: Fundamental Laws of Logarithms

1 Evaluate $\log_3(54) - \log_3(2)$ without a calculator.

a. 3

b. 2

c. 4

d. $\log_3(52)$

2 Express $2\log_a(x) - 3\log_a(y) + \log_a(z)$ as a single logarithm.

a. $\log_a\left(\frac{x^2z}{y^3}\right)$

b. $\log_a\left(\frac{2xz}{3y}\right)$

c. $\log_a(x^2 - y^3 + z)$

d. $\log_a\left(\frac{x^2}{y^3z}\right)$

3 Given $\log_b(2) = p$ and $\log_b(3) = q$, express $\log_b(72)$ in terms of p and q.

a. $3p + 2q$

b. $2p + 3q$

c. $6pq$

d. $3p - 2q$

Section 2: Solving Logarithmic Equations

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4 Solve the equation $\log_2(x) + \log_2(x-2) = 3$ for $x > 2$.

a. $x = 4$

b. $x = 4$ and $x = -2$

c. $x = 3$

d. $x = 8$

5 Solve $\log_3(2x+1) - \log_3(x-1) = 2$.

a. $x = \frac{10}{7}$

b. $x = 2$

c. $x = \frac{7}{10}$

d. No real solution

6 Solve the base-change logarithmic equation $\log_4(x) + \log_2(x) = 6$.

a. $x = 16$

b. $x = 8$

c. $x = 64$

d. $x = 4$

Section 3: Exponential Equations and Growth Models

7 Solve $3^{2x-1} = 5^x$, giving x in terms of natural logarithms.

a. $x = \frac{\ln 3}{2\ln 3 - \ln 5}$

b. $x = \frac{\ln 5}{\ln 3}$

c. $x = \frac{\ln 3}{\ln 5 - 2\ln 3}$

d. $x = \frac{2\ln 3}{\ln 5}$

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8 A bacterial culture grows according to $N(t) = 500e^{0.04t}$, where t is in hours. After how many hours will the population reach 2000?

a. $t = 25\ln 4 \approx 34.7$ hr

b. $t = 50$ hr

c. $t = 20\ln 2 \approx 13.9$ hr

d. $t = 100$ hr

9 Solve the simultaneous exponential system $2^x 4^y = 32$ and $3^{x-y} = 27$.

a. $x = 4, y = \frac{1}{2}$

b. $x = 3, y = 1$

c. $x = 5, y = 2$

d. $x = 2, y = 1$

Section 4: Reduction to Linear Form ($Y = mX + c$)

10 The variables x and y are related by $y = ax^b$. When $\ln y$ is plotted against $\ln x$, a straight line passes through $(0, 2)$ with slope 3. Find a and b.

a. $a = e^2, \quad b = 3$

b. $a = 2, \quad b = 3$

c. $a = 3, \quad b = 2$

d. $a = e^3, \quad b = e^2$

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11 The relation $y = Ab^x$ is linearized by plotting $\lg y$ against x. If the straight line has y-intercept 1 and gradient 0.3, find A and b.

a. $A = 10, \quad b = 10^{0.3}$

b. $A = 1, \quad b = 0.3$

c. $A = 10^{0.3}, \quad b = 10$

d. $A = 0.3, \quad b = 10$

12 If a graph of xy against x^2 is a straight line with slope 4 and vertical intercept -5, express y in terms of x.

a. $y = 4x - \frac{5}{x}$

b. $y = 4x^2 - 5$

c. $y = 4 - \frac{5}{x^2}$

d. $y = \frac{4}{x} - 5$

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